

HRMS for Small Business – Product Requirement Document (PRD)

1. Overview

Product Name

CogniHR (working title)

Vision

Build a lightweight, AI-powered HRMS platform for small and medium businesses with simplified HR operations, automation, and analytics.

Target Customers

- Startups
- Small businesses
- IT services firms
- Clinics & hospitals
- Agencies
- Educational institutions
- Manufacturing SMEs

Core Objectives

- Simplify employee management
 - Automate attendance and leave tracking
 - Streamline payroll workflows
 - Improve employee experience
 - Enable AI-driven HR insights
 - Provide SaaS-ready multi-tenant architecture
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2. Product Goals

Business Goals

- Launch MVP within 6 weeks
- Acquire first 10 SMB customers
- Offer subscription-based SaaS model
- Build AI differentiation in SMB HR space

Technical Goals

- Modular architecture
 - API-first design
 - Cloud-native deployment
 - AI-assisted development compatibility
 - Mobile responsive UI
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3. User Roles

Role	Description
Super Admin	Platform owner
HR Admin	HR operations management
Manager	Team supervision
Employee	Self-service access
Payroll Admin	Payroll processing
Recruiter	Hiring workflows

4. Core Modules

4.1 Authentication & Security

Features

- JWT authentication
- Refresh tokens
- Role-based access control
- Password reset
- MFA (future)
- Session management

APIs

- POST /auth/login
- POST /auth/register
- POST /auth/logout
- POST /auth/refresh
- POST /auth/forgot-password

4.2 Employee Management

Features

- Employee onboarding
- Employee profiles
- Department mapping
- Designation management
- Employment history
- Document uploads
- Emergency contacts
- Skills tracking

Employee Fields

- Employee ID
- Full Name
- DOB
- Gender
- Email
- Phone
- Address
- Department
- Designation
- Manager
- Joining Date
- Employment Type
- Salary Structure
- Documents

APIs

- GET /employees
- POST /employees
- PUT /employees/{id}
- DELETE /employees/{id}

4.3 Attendance Management

Features

- Daily attendance
- Shift management
- Geo attendance (future)
- QR attendance

- Late mark calculation
- Overtime tracking
- Work from home tracking

Attendance Status

- Present
- Absent
- Half Day
- Leave
- Work From Home

APIs

- POST /attendance/checkin
 - POST /attendance/checkout
 - GET /attendance/report
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4.4 Leave Management

Features

- Leave request
- Leave approval workflow
- Leave balance tracking
- Holiday calendar
- Comp-off management

Leave Types

- Casual Leave
- Sick Leave
- Earned Leave
- Maternity Leave
- Loss of Pay

APIs

- POST /leave/request
 - PUT /leave/approve
 - GET /leave/balance
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4.5 Payroll Management

Features

- Salary structure
- Payroll processing
- Payslip generation
- Deductions
- Tax calculations
- Reimbursements

Payroll Components

- Basic
- HRA
- Allowances
- Bonus
- PF
- Tax
- Deductions

APIs

- POST /payroll/process
 - GET /payroll/payslip
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4.6 Recruitment Management

Features

- Job posting
 - Resume upload
 - Candidate tracking
 - Interview scheduling
 - Offer management
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4.7 Performance Management

Features

- Goal setting
- KPI tracking
- Performance reviews
- Feedback system

4.8 Dashboard & Analytics

HR Dashboard

- Total employees
- Attendance summary
- Leave trends
- Payroll insights
- Attrition rate

Manager Dashboard

- Team attendance
 - Leave approvals
 - Performance metrics
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4.9 Notifications

Channels

- Email
- SMS
- Push notifications
- In-app notifications

Events

- Leave approval
 - Payroll completion
 - Attendance alerts
 - Birthday reminders
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5. AI Features

AI HR Assistant

- Employee query assistant
- Policy Q&A
- Leave recommendation
- Payroll clarification

AI Analytics

- Attendance anomaly detection
- Employee sentiment analysis
- Attrition prediction

AI Document Processing

- Resume parser
 - OCR for documents
 - Auto-fill onboarding data
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6. Non-Functional Requirements

Performance

- Page load < 3 seconds
- API response < 500ms

Scalability

- Multi-tenant SaaS
- Horizontal scaling

Security

- Encrypted passwords
- HTTPS
- Audit logs
- RBAC

Availability

- 99.9% uptime target
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7. Suggested Tech Stack

Frontend

- Next.js
- React
- Tailwind CSS

- shadcn/ui

Backend

- FastAPI or NestJS
- PostgreSQL
- Redis

AI Layer

- OpenAI API
- AWS Bedrock

Hosting

- Vercel
 - AWS ECS
 - Railway
 - Render
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8. Database Design

Tables

- organizations
 - users
 - employees
 - departments
 - attendance
 - leave_requests
 - payroll
 - salary_components
 - notifications
 - audit_logs
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9. SaaS Multi-Tenant Design

Organization — Departments — Employees — Attendance — Payroll — Leave

10. UI/UX Guidelines

Design Principles

- Minimalistic
- Mobile-first
- Responsive
- Dashboard-centric
- Dark mode support

Recommended UI Libraries

- shadcn/ui
 - Radix UI
 - Heroicons
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11. DevOps & Deployment

CI/CD

- GitHub Actions
- Docker
- Kubernetes (future)

Monitoring

- Prometheus
 - Grafana
 - Sentry
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12. Recommended Development Plan

Phase 1 – MVP (6 Weeks)

Week 1

- PRD finalization
- DB schema
- Wireframes

Week 2

- Authentication
- RBAC
- Employee module

Week 3

- Attendance module
- Leave module

Week 4

- Payroll basics

Week 5

- Dashboard
- Notifications

Week 6

- AI assistant
- Beta testing

13. Vibe Coding Strategy

Recommended AI Coding Tools

- Cursor
- Windsurf
- GitHub Copilot

Prompting Strategy

Example Prompt

Build Employee Management module using:

- Next.js frontend
- FastAPI backend
- PostgreSQL database

Features:

- CRUD
- Search

- Pagination
- JWT auth
- Validation
- Audit logs

Generate:

- API routes
- DB schema
- UI screens
- Service layer

Best Practices

- Build module-by-module
- Keep prompts focused
- Maintain API contracts
- Use reusable components
- Create shared UI library

14. Future Roadmap

Mobile Apps

- Flutter app
- Employee self-service app

Advanced AI

- HR Copilot
- Voice assistant
- AI reporting

Enterprise Features

- Compliance
- Tax automation
- Face recognition attendance
- Shift planning

15. Monetization Strategy

Pricing Plans

Starter

₹999/month

- Up to 25 employees

Growth

₹2999/month

- Payroll
- AI assistant
- Up to 100 employees

Enterprise

Custom pricing

16. Competitive Advantage

Differentiators

- AI-first SMB HRMS
 - Fast onboarding
 - Healthcare workforce optimization
 - Modern UX
 - Lightweight architecture
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17. Recommended Immediate Next Steps

1. Finalize wireframes
2. Create database schema
3. Setup GitHub repositories
4. Setup CI/CD pipeline
5. Build authentication
6. Start employee module
7. Release internal beta
8. Gather SMB feedback

18. Conclusion

This HRMS platform should focus on simplicity, automation, and AI-driven operational efficiency for small and medium businesses.

The recommended approach is:

- Start lean
- Build modular
- Launch quickly
- Iterate based on feedback
- Add AI differentiation progressively